

CHAPTER 28

Development and Inheritance

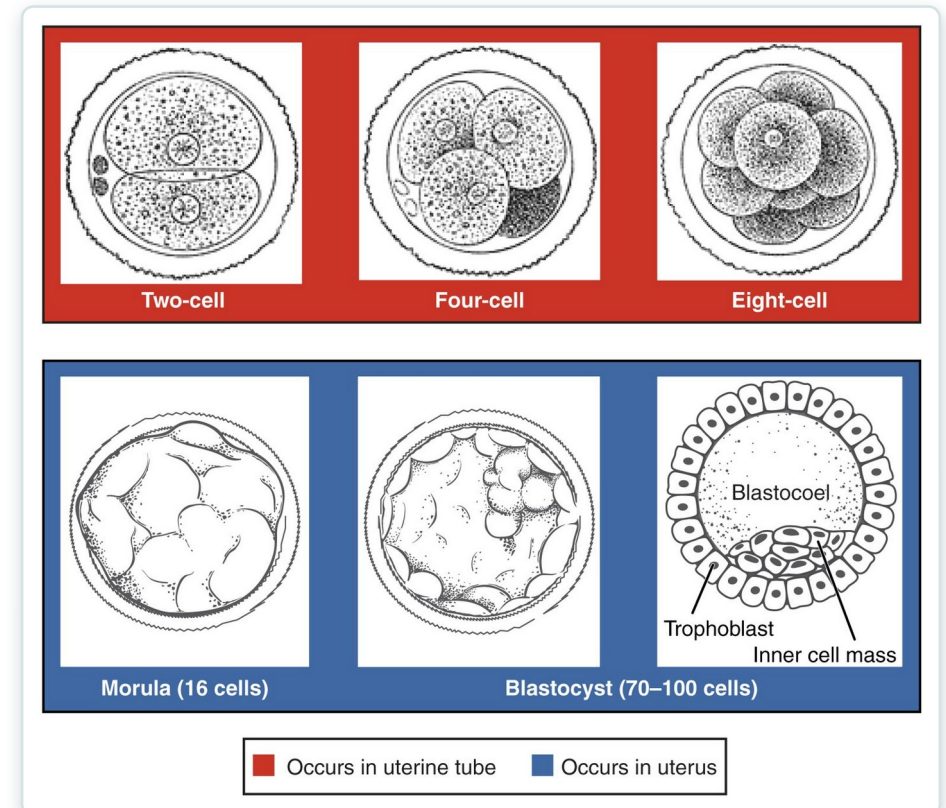
From fertilization to birth, and how traits are passed on.



Learning Objectives

After studying this chapter, students will be able to:

- 1 List and explain the steps involved in fertilization
- 2 Describe the major events in embryonic development
- 3 Describe the major events in fetal development
- 4 Discuss the adaptations of the body to pregnancy
- 5 Describe the physiologic adjustments that the newborn must make in the first hours of extrauterine life
- 6 Summarize the physiology of lactation



Early cleavage of the fertilized egg.



28.1

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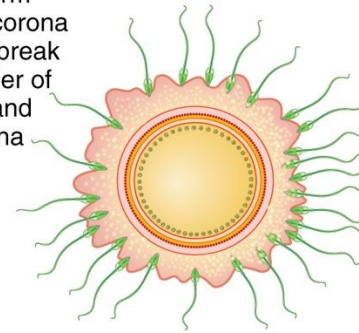
Fertilization

How sperm and egg unite to begin a new life.

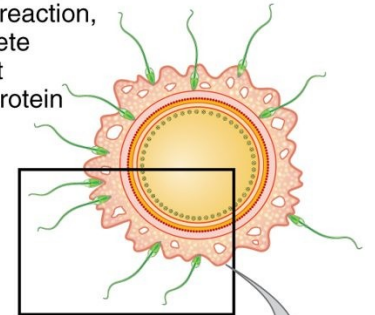
Fertilization

- Sperm meets egg in the uterine tube
- One sperm penetrates the egg
- Their chromosomes combine
- Forms a single-celled zygote

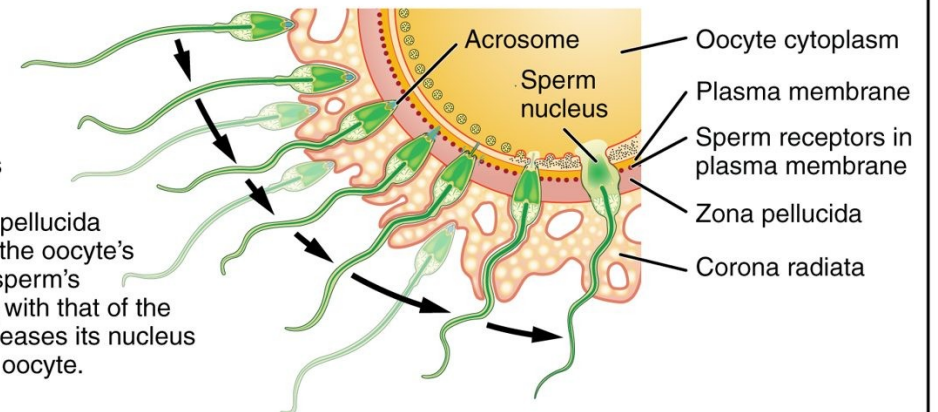
① Hundreds of sperm attracted to the corona radiata begin to break through the barrier of granulosa cells and approach the zona pellucida.



② Contact with the zona pellucida triggers the acrosome reaction, causing sperm to secrete digestive enzymes that break down the glycoprotein membrane of the zona pellucida and help to expose the oocyte's plasma membrane.



③ A single sperm succeeds in burrowing through the corona radiata and zona pellucida and making contact with the oocyte's plasma membrane. The sperm's plasma membrane fuses with that of the oocyte and the sperm releases its nucleus into the cytoplasm of the oocyte.



Fertilization



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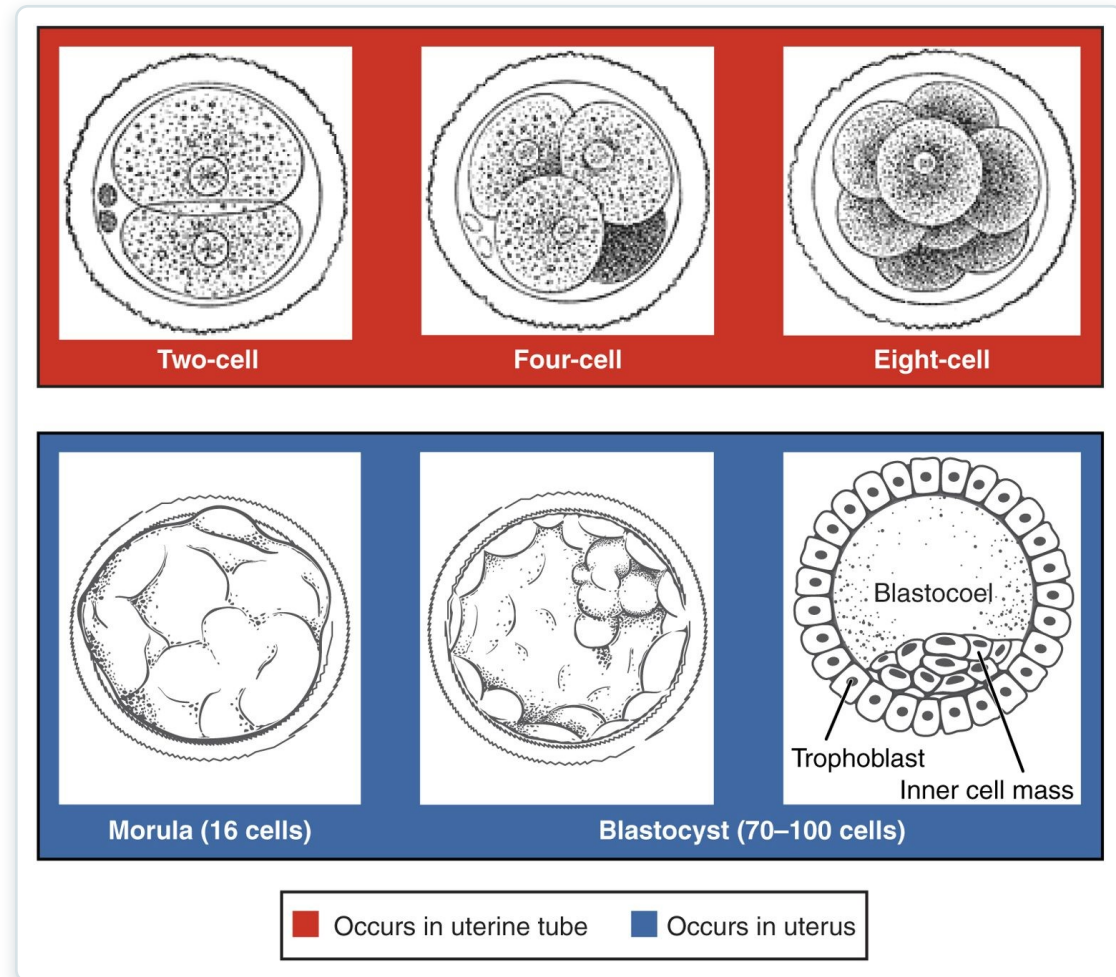
Embryonic Development

From a single cell to an implanted embryo with three germ layers.

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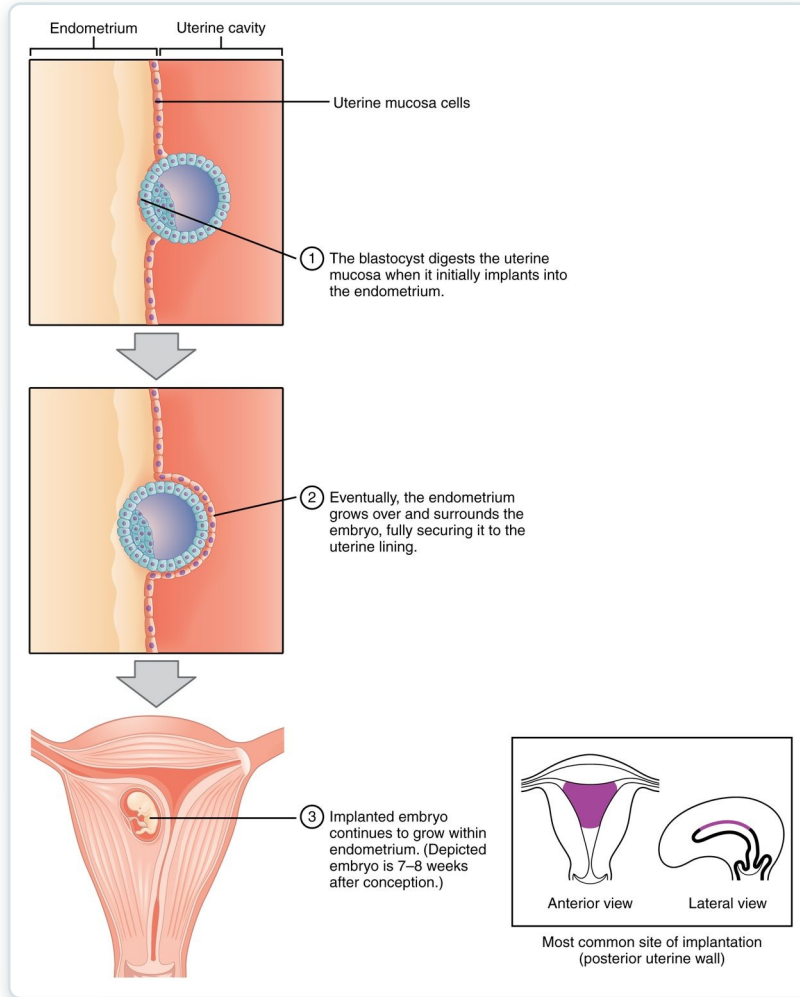
Cleavage

- The zygote divides repeatedly
- 2-cell, 4-cell, 8-cell stages
- Becomes a solid morula
- Then a hollow blastocyst



Cleavage of the zygote

Implantation



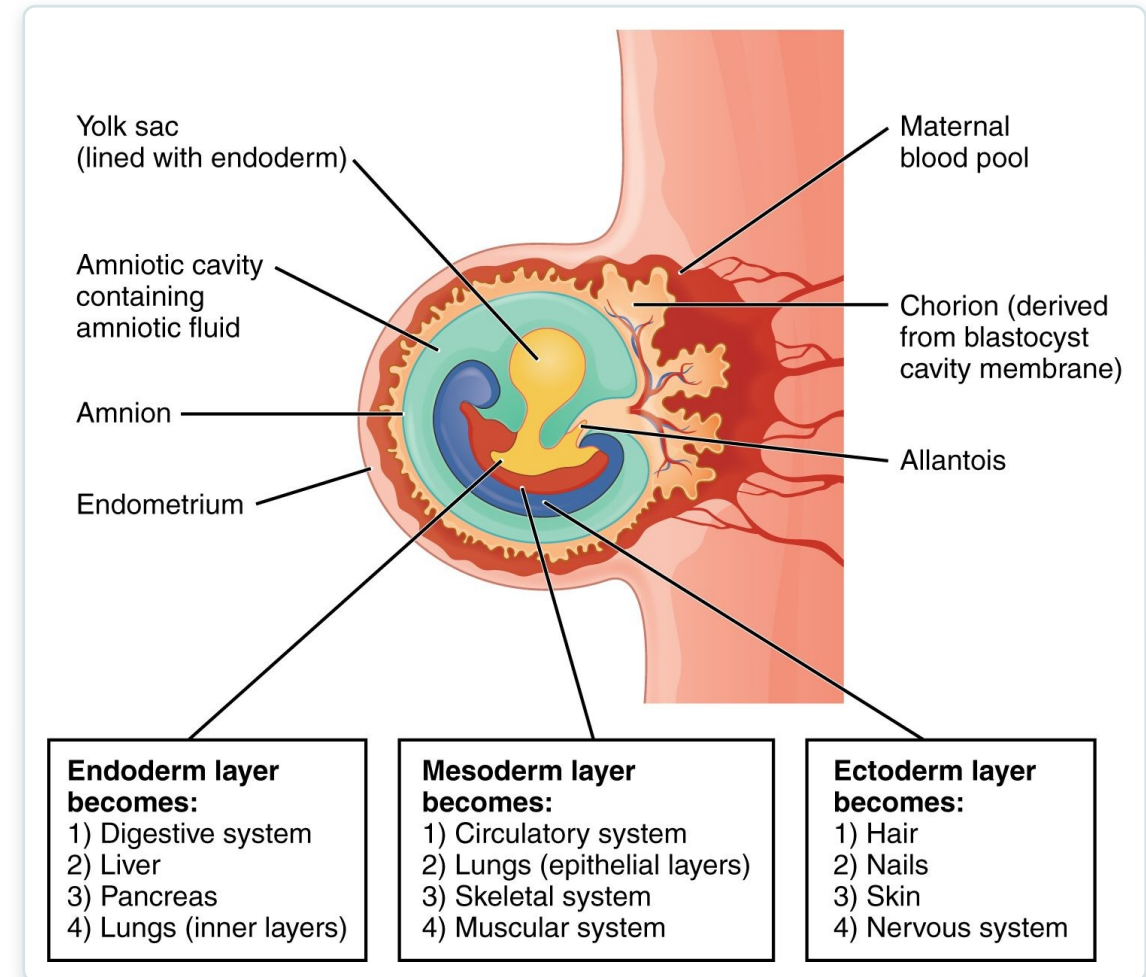
Implantation in the uterus

- The blastocyst reaches the uterus
- It embeds in the uterine lining
- This is implantation
- It establishes the pregnancy

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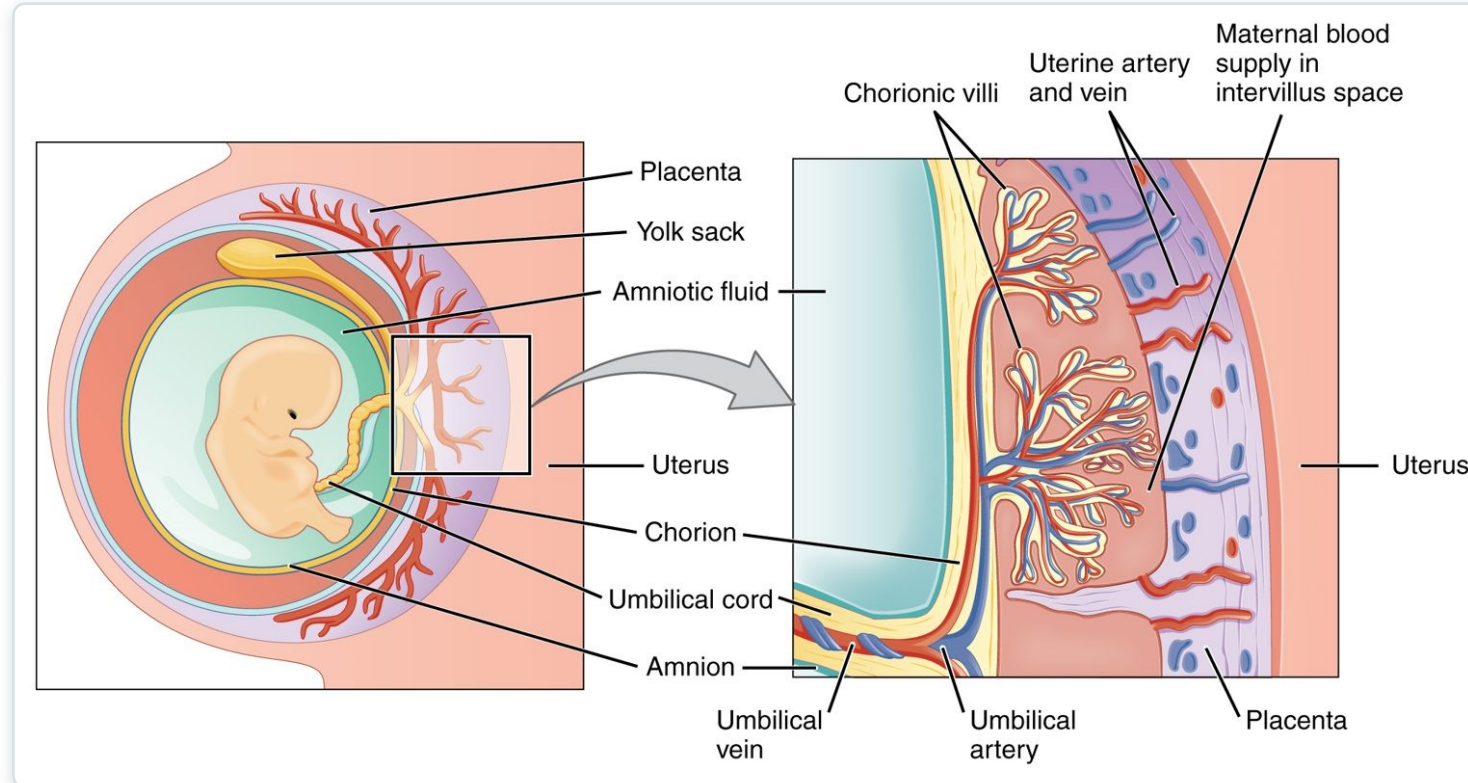
The Germ Layers

- The embryo forms three layers
- Ectoderm: skin and nervous system
- Mesoderm: muscle, bone, heart
- Endoderm: gut and glands



The three primary germ layers

The Placenta



The placenta

- The lifeline between mother and fetus
- Exchanges oxygen and nutrients
- Forms from embryo and uterus
- Removes fetal wastes



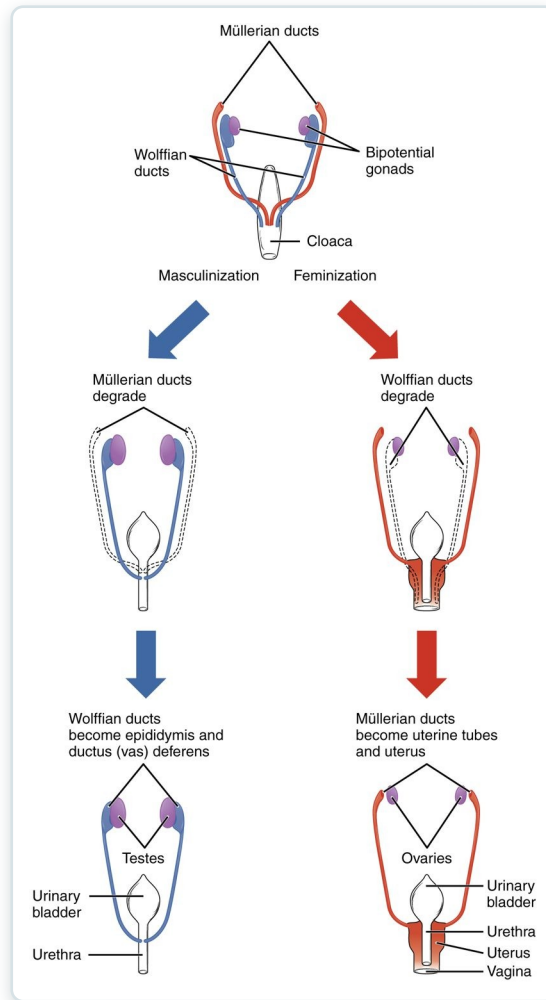
28.3

SECTION

Fetal Development

The growth and maturation of the fetus.

Sexual Differentiation



Development of the reproductive structures

- Early structures are the same
- Sex develops over the fetal period
- Determined by the sex chromosomes
- Male or female structures form



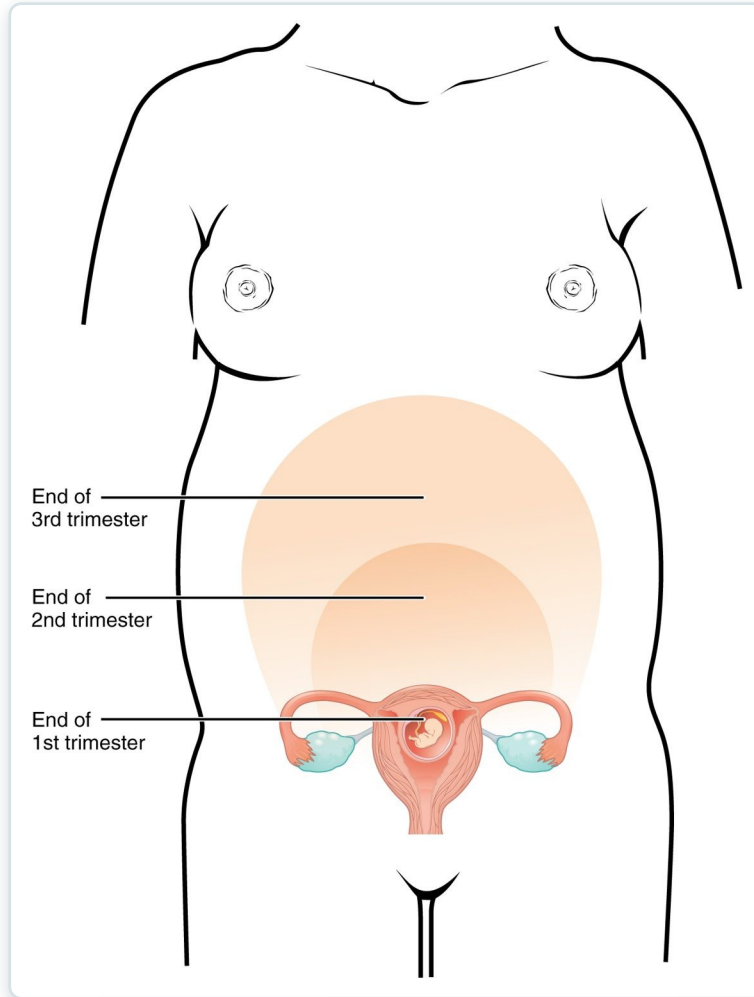
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SECTION

Changes During Pregnancy, Labor, and Birth

How the mother's body changes through pregnancy and birth.

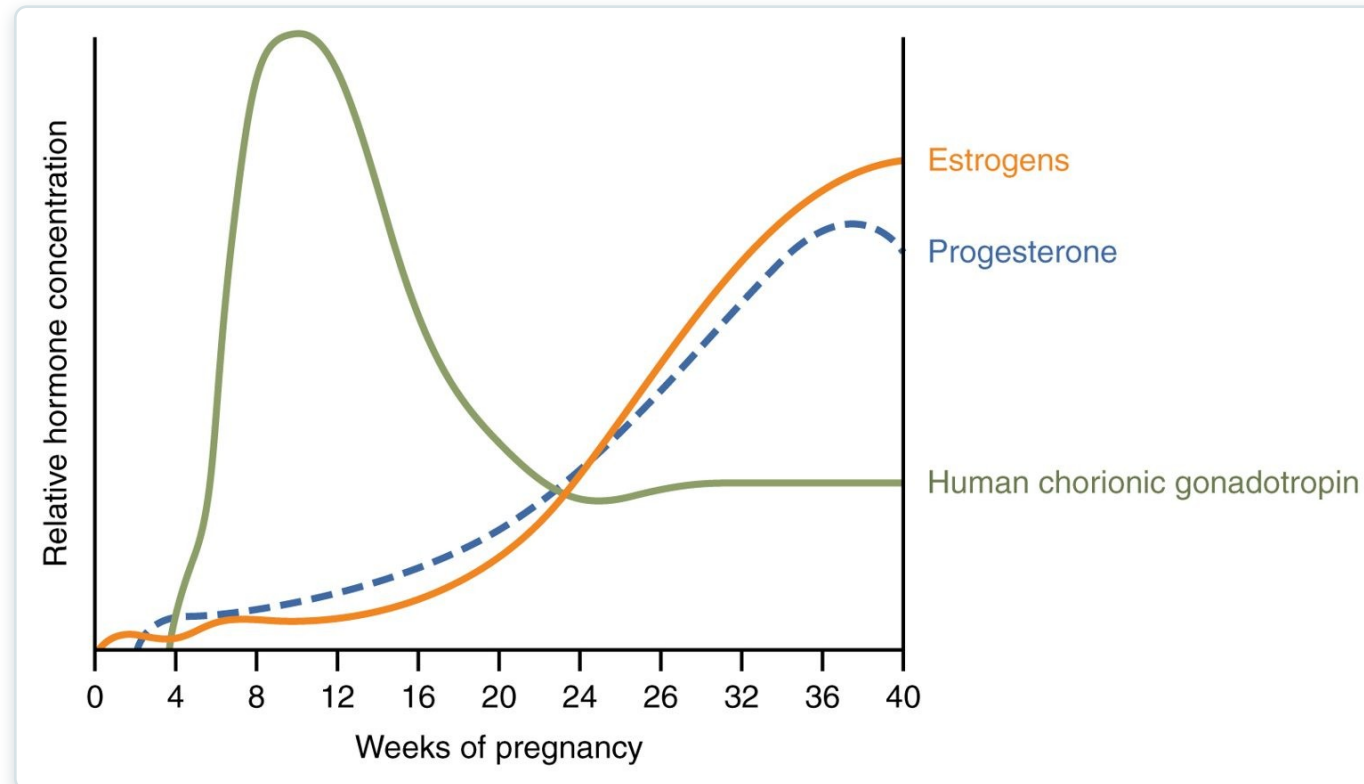
Changes During Pregnancy



Maternal changes by trimester

- The body adapts to support the fetus
- The uterus expands greatly
- Trimesters mark the stages
- Many systems adjust

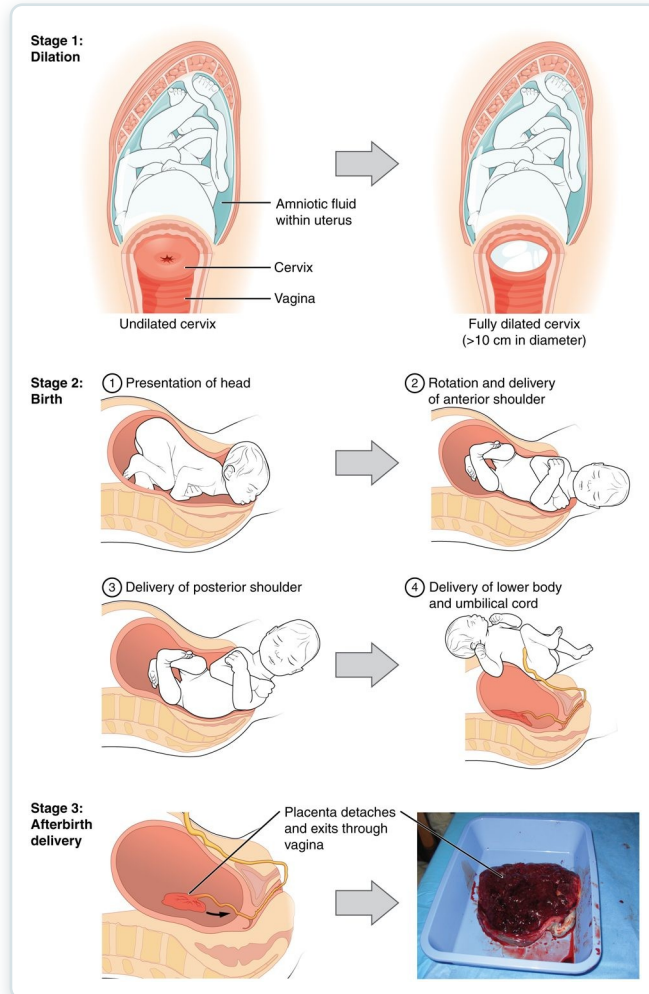
Hormones of Pregnancy



Hormone levels during pregnancy

- Hormones maintain the pregnancy
- hCG signals early pregnancy
- Estrogen and progesterone rise
- hCG is what pregnancy tests detect

Labor & Birth



The stages of labor

- Labor delivers the baby
- Stage 1: the cervix dilates
- Stage 2: the baby is born
- Stage 3: the placenta is delivered



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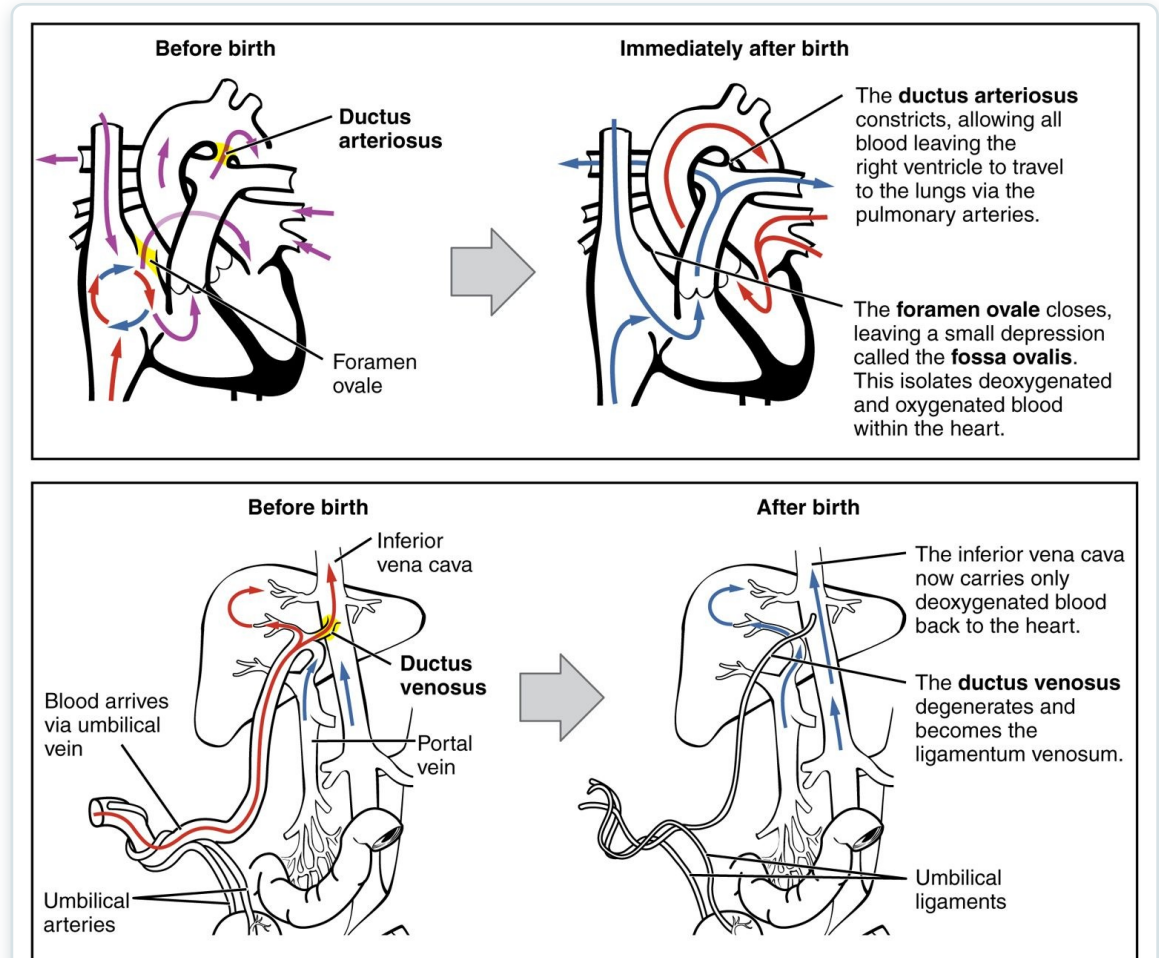
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Adjustments of the Infant at Birth and Postnatal Stages

How the newborn's body adjusts to life outside the womb.

Adjustments at Birth

- Big changes at the first breath
- The lungs inflate and take over
- Fetal circulation shunts close
- The newborn becomes independent



Circulatory changes at birth



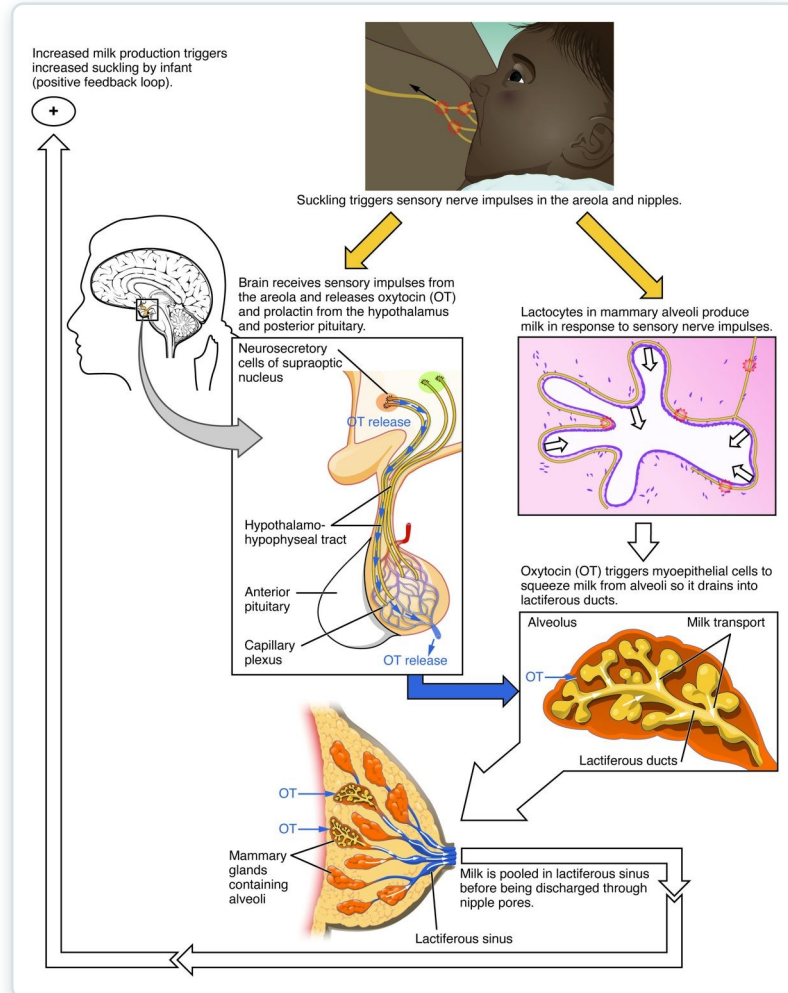
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SECTION

Lactation

How the breasts produce milk to nourish the newborn.

Lactation



Lactation

- The breasts make milk after birth
- Prolactin drives milk production
- Oxytocin triggers milk letdown
- Suckling sustains the supply

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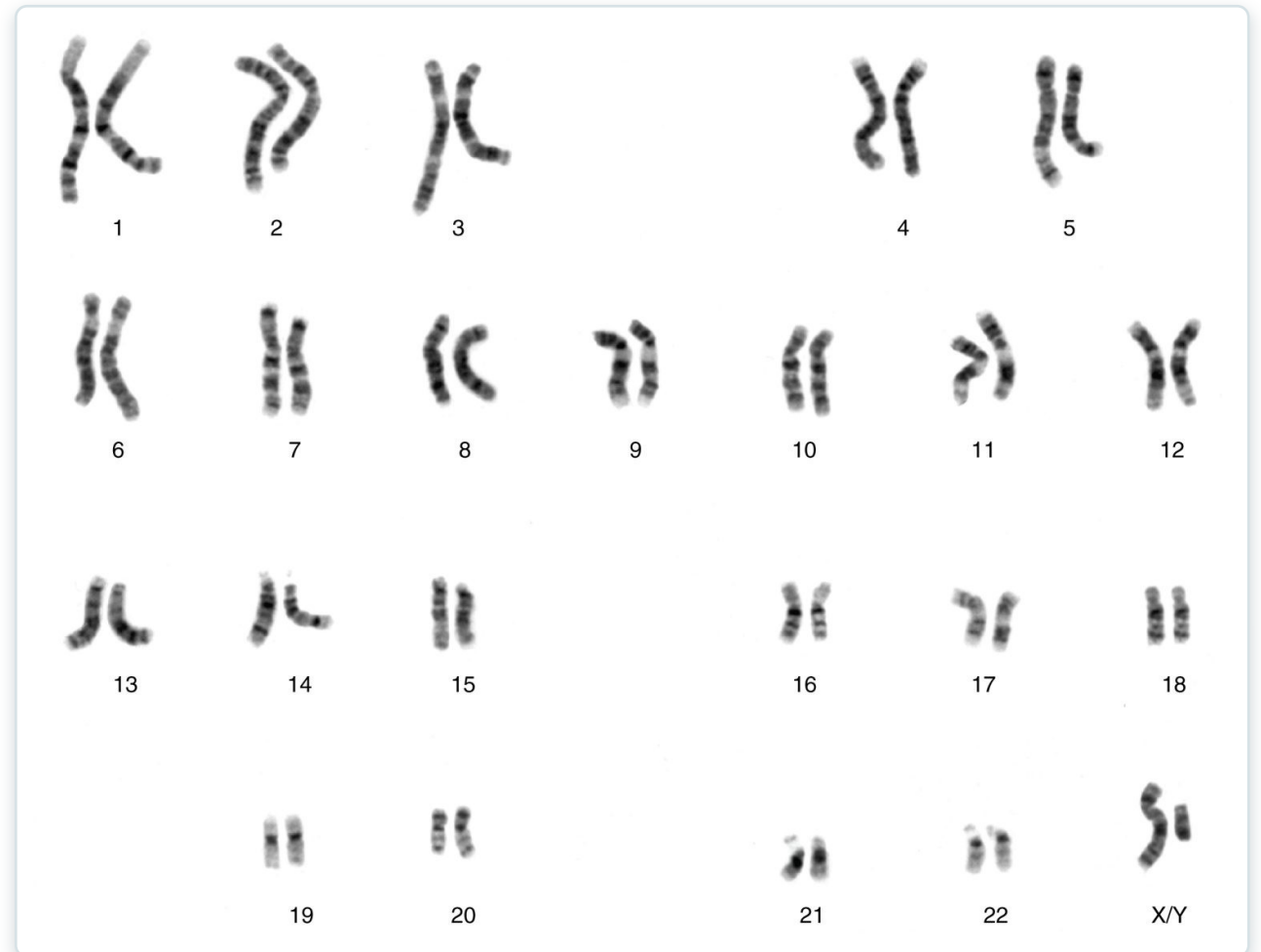
SECTION

Patterns of Inheritance

How traits are passed from parents to children.

The Karyotype

- Traits are carried on chromosomes
- 46 chromosomes in 23 pairs
- 22 pairs plus the sex chromosomes
- XX is female, XY is male

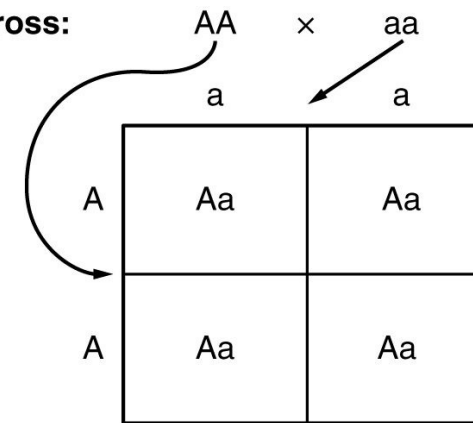


A human karyotype

Patterns of Inheritance

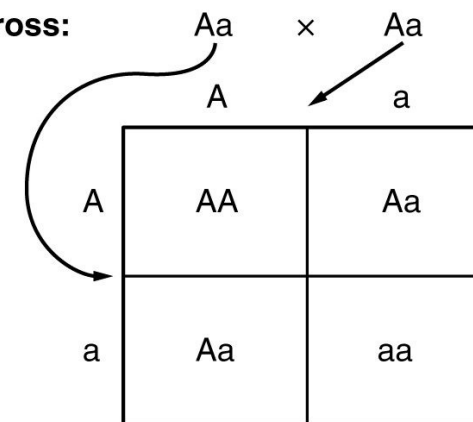
- Genes come in alternate forms (alleles)
- Dominant alleles mask recessive ones
- A Punnett square predicts offspring
- Explains many inherited traits

Generation 1 Cross:



Generation 2:
100% of offspring
are Aa (dominant)

Generation 2 Cross:

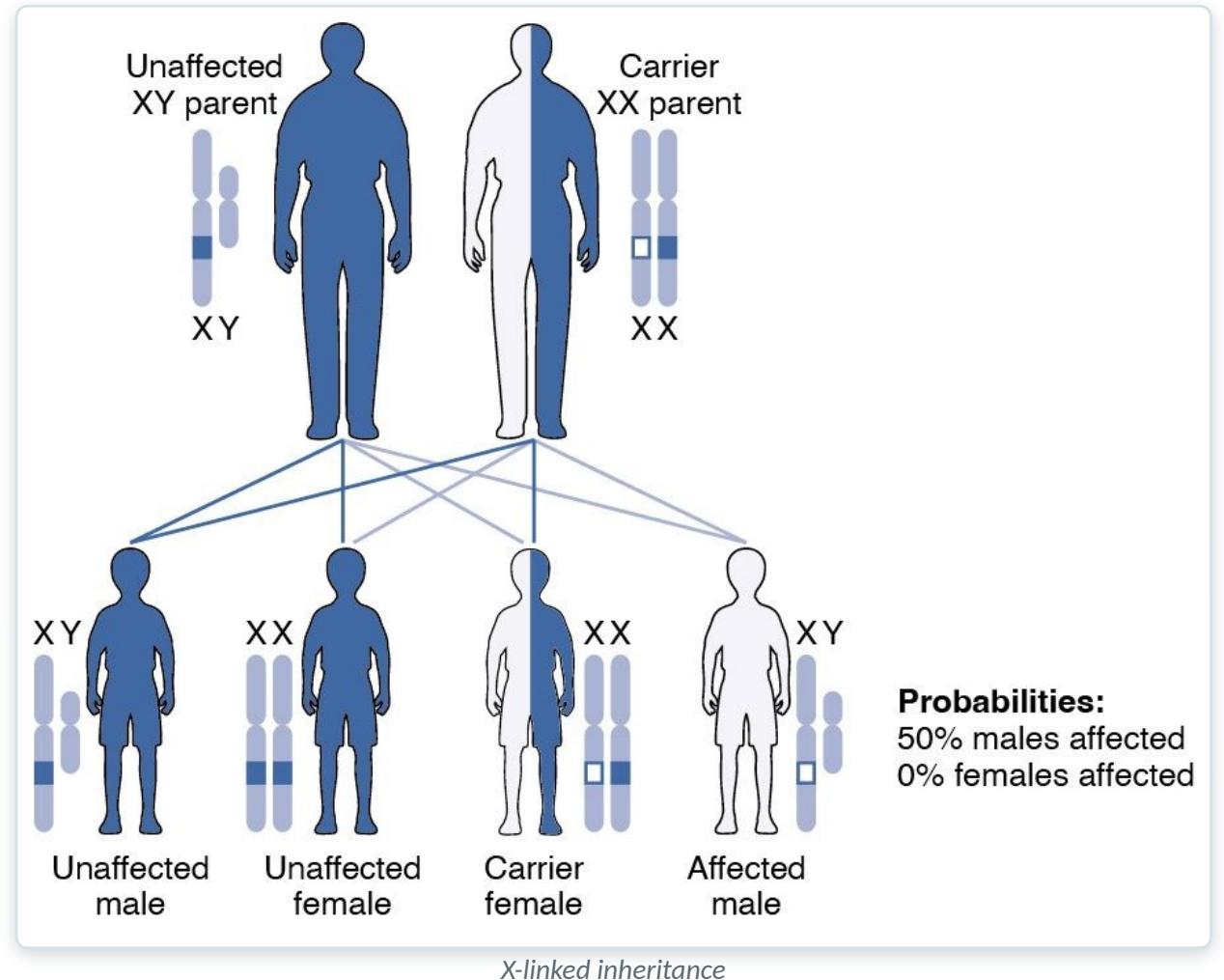


Generation 3:
75% AA or Aa
(dominant)
25% aa recessive

A Punnett square

Sex-Linked Inheritance

- Some genes sit on the X chromosome
- These show different patterns
- More common in males
- Examples: hemophilia, color blindness



Key Takeaways

- Fertilization unites sperm and egg into a zygote with a full set of chromosomes.
- The embryo develops through cleavage, implantation, and the three germ layers.
- The placenta exchanges nutrients, oxygen, and wastes between mother and fetus.
- Labor delivers the baby in three stages, and the newborn adjusts to breathing on its own.
- Traits are inherited through chromosomes and genes following predictable patterns.

Why Development & Inheritance Matter in Practical Nursing

Prenatal Care

Monitoring supports a healthy pregnancy and fetus.

Labor & Delivery

Understanding the stages guides safe nursing care.

Newborn Assessment

The first checks confirm a healthy transition.

Breastfeeding

Supports infant nutrition and immunity.

Genetic Disorders

Inherited and chromosomal conditions affect care.

Congenital Conditions

Some problems arise during development.